KAI WICKE

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First-year Mechanical Engineering student at Northeastern University with a background in retail and teaching. Deeply interested in how CAD, robotics, and AI can be used to build real-world technology—from competing in Maker Faire competitions to building AI-powered tools. Strong interest in mathematics, golf, weightlifting, chess, and teaching others about the things I'm passionate about.

EDUCATION

Northeastern University 2025 – 2029
Bachelor of Engineering — Mechanical Engineering | London / Boston
Relevant coursework: Physics, Calculus III, Differential Equations, Engineering Design, Chemistry

St. Mary's College High School 2021 – 2025
High School Diploma | Berkeley, CA

- 3.88 GPA | 6 Advanced Placement courses (STEM and Humanities)
- Spirit and Athletic Director (elected Student Leadership position)
- Varsity Golf (4 years, Coach Rau — home course: Tilden) | Varsity Soccer
- National Honor Society member

EXPERIENCE

Summer Camp Counselor May 2023 – Jul 2025
JCC East Bay | Oakland, CA

- Led games, activities, and science experiments for groups of 10–20 campers aged 5–12
- Provided inclusive support for children with disabilities, ensuring a safe and enjoyable experience
- Developed communication, conflict resolution, group management, and event planning skills

Retail Salesperson Dec 2024 – May 2025
Ole Factorie Candles | Oakland, CA

- Assisted customers, processed sales, and crafted handmade candle products in a local retail shop
- Built customer relationships through attentive service and product recommendations

Mathematics & Chemistry Tutor Sep 2024 – May 2025
Self-Employed | Oakland, CA

- Tutored high school students in trigonometry and chemistry, tailoring explanations to individual learning styles
- Strengthened mentoring, patience, and the ability to break down complex concepts clearly

Counselor in Training Jun 2022 – Aug 2022
Nike Sports Camps

- Organized activities and supplies for campers aged 5–12; independently led groups of up to 20
- Gained early experience in leadership, planning, and group management

PROJECTS

FocusFinder — Attention-Tracking Desk Robot 2026

- Built a desk-mounted productivity tool using a Raspberry Pi Zero 2W with OpenCV face detection to monitor user attention in real time

- Animated robot eyes on a transparent OLED follow the user; a Pico-controlled LED and buzzer system provides passive physical feedback when focus is lost

SwingIQ — AI Golf Swing Analyzer 2026

- Developed a web application using Google Gemini AI and computer vision to analyze golf swing form, track metrics, and suggest personalized drills
- Built with React and vanilla JavaScript; deployed via Netlify (GitHub: kaiwicke07)

Autonomous Robotic Model Car 2025

- Designed and built a sensor-equipped robot capable of autonomous driving with wall detection and obstacle avoidance

Maker Faire — Catapult Build 2020

- Designed and constructed a functional catapult for a regional Maker Faire competition, applying prototyping and physics principles

SKILLS

Technical: Python, C++, HTML/CSS | CAD (SolidWorks, Fusion 360) | Robotics | 3D Printing & Prototyping | Circuit Design | OpenCV

Interpersonal: Customer service | Sales | Communication | Active listening | Conflict resolution | Adaptability | Patience | Teamwork | Reliability | Time management | Leadership | Mentoring | Public speaking | Event planning | Inclusivity

INTERESTS

Golf | Weightlifting | Chess | Hiking & Camping | Robotics | Mathematics